REFLOWECONOMY · CONCEPT OPTIONS

ReflowEconomy concept options

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First concept options at the kit 1.9.1 upgrade; the existing reference micro-factory is option A; option B (sort-and-bale hub) added and modelled at D3; scored against RFE-URS-001; choice proposed, awaiting Amish

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Recommendation: the sort-and-bale hub (B) scores higher than the reference micro-factory (A), 228 against 180 of 295, so the choice goes to Amish. The recommendation is to keep A as the playbook's reference design, because B does not do what the pitch promises (recover and remanufacture locally) and scores 1 of 5 on keeping value local, and to graft B's best idea onto A: a site can open as a sort-and-bale hub on a single-phase supply and add the wash, shred and melt line once its supply and buyers are confirmed. Proposed, awaiting Amish (RFE-DEC-001 D4, RFE-DDR-004). Method: STANDARDS section 21.

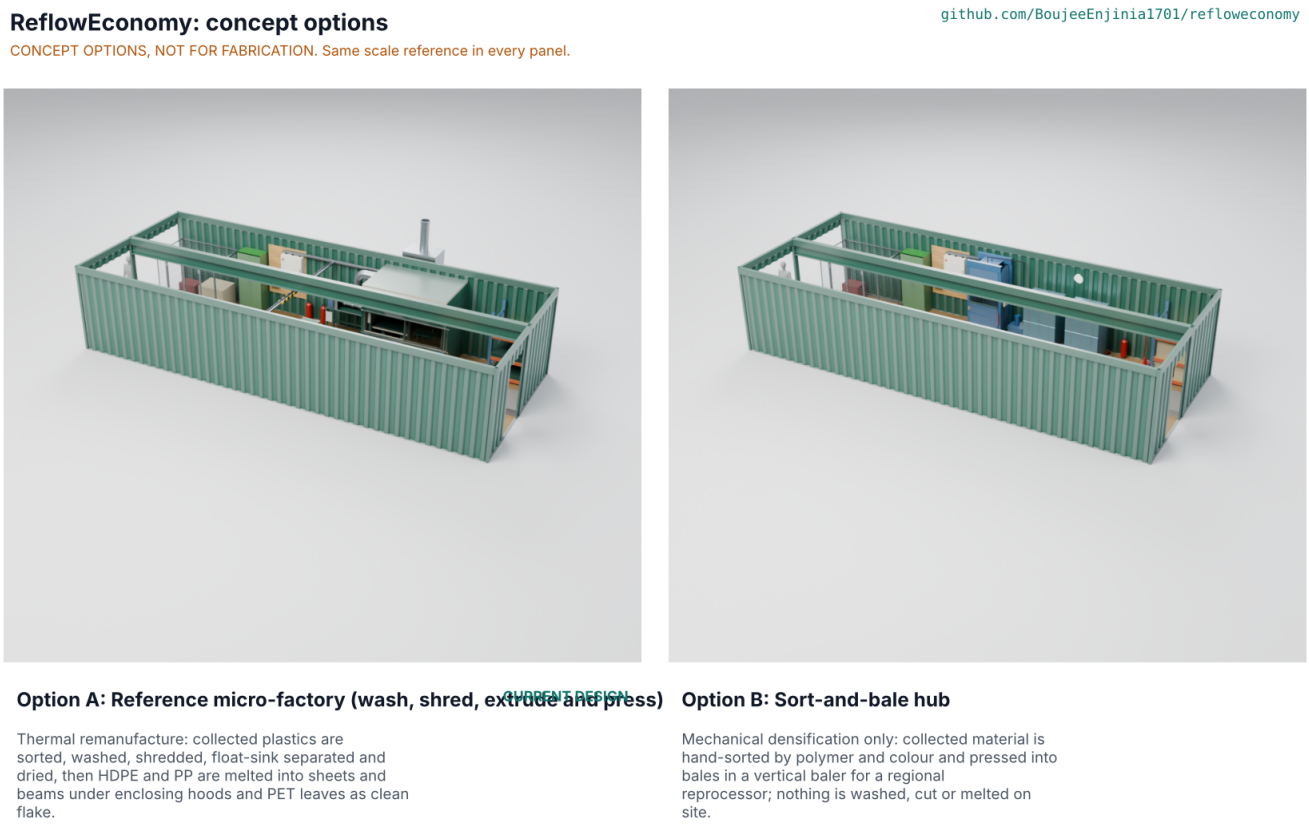


Figure 1. The two options at one scale, rendered in Blender Cycles from `components()` *in* `cad/src/model.py` *by* `python .kit/options.py` *(rendered on Amish's machine).*

The options

Option A: reference micro-factory (wash, shred, extrude and press)

Working principle: thermal remanufacture. Collected plastics are hand-sorted, washed, shredded, float-sink separated and dried; HDPE and PP are melted into 1 x 1 m sheets and beams under two enclosing hoods, and PET leaves as clean flake. This is the existing design (RFE-PRC-001, RFE-CAL-001, RFE-BLD-001), unchanged except for the 120 mm feed slot fixed under the STEM check (RFE-DEC-001 D3).

On paper it takes 100 kg of collected input a shift and keeps 51.6 % local as products and flake, sends 28.4 % to licensed disposal and uses 0.88 kWh per kilogram of output. Its equipment is USD 25,505 (indicative, USD 505 over the value-engineering target). Counted as currents, its morning loads draw about 55 A from a supply decided as 230

V, 40 A single-phase (RFE-STM-001 S14; RFE-DEC-001 D1). Key risks: the supply; throughput margins on the shredder and the derated press; fumes and exposure, which only air monitoring can confirm; the product price (the site breaks even at 2.49 USD/kg of product, while regenerated HDPE sheet is listed in India at about 0.5 to 1 USD/kg, RFE-MKT-001). From a co-design partner it needs the input mix, local prices and buyers for sheets and beams, and the supply a site can get.

Option B: sort-and-bale hub

Working principle: mechanical densification only. Collected material is weighed, hand-sorted by polymer and colour at two sorting tables into eight bins and three bulk bags, and pressed into bales of about 120 kg in a 10 t electro-hydraulic vertical baler; bales go to a regional reprocessor with a passport. Nothing is washed, shredded or melted, so there is no wet area, no hot zone and no fume extraction. Modelled at D3 in the same container shell (`cad/src/detail_b.py`): scale, two tables, eight bins, three bulk-bag stands, the baler with its power pack, two pallets of bales, a pallet truck, the export cage, safety station, board, racking and desk.

First-order numbers: three sorters at the 25 kg/h of RFE-CAL-001 for 4 h and a fourth person weighing, baling and issuing passports handle about 300 kg a shift (RFE-STM-001 B4). Per 100 kg of input, 60 kg of sorted plastic, 8 kg of metals and 10 kg of paper are baled for sale, 2 kg of e-waste is exported and 20 kg of residue goes to disposal; none is remanufactured on site. The baler, lights and a fan need about 24 A (B3). Key risks: bale prices (unwashed PET bottle scrap is listed in India at about INR 55/kg, about 0.58 USD/kg; HDPE and PP bale prices to assess), the value leaves for regional reprocessing, a high fire load of stored bales, and the crush hazard of the baler (interlocked loading gate). From a co-design partner it needs the reprocessors who would buy bales, their grade rules and prices.

Screened out before modelling

A third principle was screened and not modelled: melt-mixing shredded plastic with sand into paving blocks. It is a melt route like A, with the fume hazard of mixing in open or semi-open kettles, an abrasive product that sheds plastic as it wears, and a single low-value product, so it adds nothing A does not already test.

First-order numbers

Table 1. Options compared (model figures from `cad/fidelity.json`; costs indicative).

Quantity	Option A	Option B
Estimated equipment cost (USD)	25,505 (bom/bom.csv items 1 to 12)	About 8,000: A's lines 1, 8, 9, 10, 11 and 12 at their indicative prices (USD 6,450) plus a 10 t vertical baler listed at INR 150,000 (about USD 1,570 at INR 95.46 per USD); second table, bag stands and pallet truck to assess
Model mass, all parts (kg)	9,779	7,685
Input per 8 h shift	100 kg	About 300 kg (sorting-limited)
Output per 100 kg of input	32.3 kg products, 19.3 kg clean flake; 28.4 kg to disposal	60 kg sorted baled plastic, 18 kg baled metals and paper; 20 kg to disposal
Kept local as products or clean feedstock	51.6 %	0 % (bales go to a regional reprocessor)
Energy	45.6 kWh per shift, 0.88 kWh/kg of output	About 9 kWh per shift (baler about 2 h of cycles, lights, fan), under 0.05 kWh/kg
Supply current, morning block	About 55 A running on a 40 A single-phase supply	About 24 A
Parts in the D3 model	1,005 (294 fasteners, 168 welds)	605 (153 fasteners, 92 welds)
STEM check	Open: S14 supply (to Amish), S15 noise and S21 exposure (site data)	Within known science; B1 bale density to confirm

Scoring

Table 2. Weighted scoring against RFE-URS-001 (weights: High UR 3, Medium 2; scores 1 to 5).

Criterion	Source	Weight	A	B	Reason
What to process here	UR-001	3	4	4	The feasibility matrix serves both
Recipes we can follow	UR-002	3	4	4	A has four process recipes; B needs sorting and baling rules only
Know if it pays	UR-003	3	3	3	Both use the economics model; A breaks even only at its assumed product price, B's bale prices are to assess
Send away safely	UR-004	2	3	3	Same export cage and rules in both
Fit a small site	UR-005	3	4	5	A fits with its hot zone at the 1.0 m limit; B fits with room to spare
Handle a day's collection	UR-006	3	3	5	A is at risk (shredder 5 % margin); B handles about 300 kg
Keep value local	UR-007	3	4	1	A keeps 51.6 % local as products and flake; B adds only sorting value and ships bales away
Little to disposal, nothing burned	UR-008	3	2	3	A 28.4 % with process losses; B 20 %, the sorting residue alone
Low electricity bill	UR-009	2	3	5	0.88 against under 0.05 kWh/kg
Use the supply we can get	UR-010	3	1	5	A needs about 55 A on paper against 40 A; B about 24 A
Clean air	UR-011	3	3	5	A melts plastic under hoods (to verify); B melts nothing
Safe machines	UR-012	3	3	4	A has a shredder, hot presses and noise; B has one interlocked baler
Trusted material	UR-013	3	4	3	Passports on both; A's flake and products carry more process data
Quick records	UR-014	2	3	3	Same desk and passport
Build and repair locally	UR-015	2	3	4	A needs three open machines built by a capable workshop; B a bought baler and simple frames
Affordable equipment	UR-016	2	2	4	USD 25,505 against about USD 8,000
Permittable	UR-017	3	3	4	Both register as processors; A also needs an air consent for melting (to assess)
Fire safety	UR-018	3	3	4	A has a hot zone beside plastic stock; B a high fire load of bales but no heat
Fair intake rules	UR-019	2	3	3	Same intake rule
Cost against the value-engineering target	PR-010	2	3	5	A USD 505 over the USD 25,000 target; B well under
Buildable with local skills and tools	BLD	2	3	4	A's booth, enclosure and ducting against B's frames and a bought baler
Safety, overall hazard profile	RSK	3	3	4	Fewer and lower hazards in B

Criterion	Source	Weight	A	B	Reason
Maintenance	RSK	1	2	4	A has filters, heaters, cutters and three drives; B one hydraulic pack
Weighted total			**180**	**228**	Of 295

Recommendation

B outscores A by more than 10 %, so this goes to Amish as a decision (STANDARDS section 21; the repo is not redesigned without his go). The scores favour B because it uses little power, needs little capital and has few hazards; most of A's lost points come from its supply problem (RFE-DEC-001 D1) and its thin economics. But B does not meet the pitch: it recovers and sorts but does not remanufacture, and the value leaves the community as bales (UR-007 scores 1 of 5). The recommendation is to keep A as the reference micro-factory and to graft B's best idea onto it as a staged start: the playbook describes how a site opens as a sort-and-bale hub on single phase (B's tables, bins, bag stands and baler fit A's front row and cage), earns while it finds buyers and the right supply, and then adds the wash, shred and melt line. A change of reference to B, or the staged start, changes the playbook's scope and is Amish's decision (RFE-DEC-001 D4, argued in RFE-DDR-004).